

Finding the Inverse of some Shift Invariant Transformations

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Abstract. We study the problem of how to find the inverse of shift invariant (SI) transformations proposed in Daemen’s thesis. In particular, two of them have been used in practice: $y_i = x_i \oplus \overline{x_{i+1}}x_{i+2}$ and $y_i = x_i \oplus \overline{x_{i+1}}x_{i+2}x_{i+3}$. The first one is the well-known χ transformation used in **SHA-3**, **Subterranean 2.0** and **Rasta**, while the second one is used in a recently proposed ZK-friendly hash function called **Monolith**. While the concrete formula of the inverse of χ of arbitrary size has been given and proved by Liu et al. at JoC 2022, it remains unknown how to deduce such a formula and how to systematically study other SI transformations. In this work, we aim to provide a general method and flow to find the inverse of SI transformations, though it is still limited to some specific types and it may not work for all such transformations. However, such a general method does shed new insight on how to find their inverse, as we can apply this method to several different SI transformations, including the one used in **Monolith**. We expect that this method can be further generalized and applied to more SI transformations.

Keywords: Boolean functions, Chi, Cryptography, Topical collection on advances in Boolean functions with applications in cryptography.

1 Background

The well-known n -bit shift invariant transformation $\chi_n : \mathbb{F}_2^n \mapsto \mathbb{F}_2^n$ is defined as follows:

$$\forall i \in [0, n-1] : y_i = x_i \oplus \overline{x_{i+1}}x_{i+2}, \quad (1)$$

where the indices are within modulo n , and

$$x = (x_0, x_1, \dots, x_{n-1}), \quad y = (y_0, y_1, \dots, y_{n-1})$$

are the n -bit input and output of χ_n , respectively.

The χ_n transformation was first proposed as a concrete and efficient nonlinear SI transformation in Daemen’s thesis. Later, χ_n has been used in several famous symmetric-key primitives like **SHA-3** [5], **Subterranean 2.0** [3,6] and **Rasta** [4]. Especially for **Rasta**, its security is closely related to the high algebraic degree of the inverse of χ_n , though no concrete formula of it was known when **Rasta** was proposed. To better understand the security of **Rasta**, Liu et al. were motivated to rigorously study the formula of the inverse of χ_n for the first time. At JoC 2022 [10], they proved that the inverse of χ_n has an elegant formula, as shown below:

$$\forall i \in [0, n-1] : x_i = y_i \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2j+1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k}}, \quad (2)$$

where the indices are within modulo n . At the same time, some algebraic properties were also studied by Schoone and Daemen [11,12]. Since then, there are several papers published to study the properties of χ_n [7,9].

In this work, instead of studying other algebraic properties of χ_n , we continue the study on the inverse of χ_n . More specifically, it remains unknown how Equation 2 was deduced, and whether it is possible to find the inverse of other SI transformations in Daemen’s thesis is also unclear. Although there are many such SI transformations in [2], as far as we know, only two are practically used to construct symmetric-key primitives. One is the above χ_n transformation, and the other is

$$\forall i \in [0, n-1] : y_i = x_i \oplus \overline{x_{i+1}}x_{i+2}x_{i+3},$$

where the indices are within modulo n . This SI transformation has been adopted in a recently proposed ZK-friendly hash function called **Monolith** [8]. For convenience, we denote this SI transformation by ψ_n .

Our contributions. First, we observe that the proof of the correctness of Equation 2 is quite long and tedious in [10], and how Equation 2 was found is kept secret. As the first contribution, we show how to deduce the inverse of χ_n and directly prove its correctness at the deduction phase. Such a proof is further generalized, and it is also applied to other SI transformations, covering ψ_n used in **Monolith**, and two SI transformations named ζ_n and ϕ_n . Although we cannot give a general method that can work for any SI transformations, we believe the proposed flow to find the inverse can inspire future study on more SI transformations.

Technical overview. Before moving to the technical details of how to derive the formula of these SI transformations, we first give a high-level description of our approach and the flow to find the inverse.

Let us take the χ_n function as an example, and this also holds for the remaining SI transformations studied in this paper. To find the inverse, our goal is to find n polynomials $q_0, \dots, q_{n-1}$ in $y_0, \dots, y_{n-1}$ such that

$$\forall i, j \in [0, n-1] : x_i y_j = q_i y_j.$$

Then, we prove that the inverse is

$$\forall i \in [0, n-1] : x_i = q_i.$$

Note that since y_j can be zero, $x_i y_j = q_i y_j$ does not necessarily imply $x_i = q_i$. Hence, we will address this issue after finding such $q_0, \dots, q_{n-1}$, and this is always the last step of our approach.

Moreover, according to Daemen's thesis [2], the inverse of a SI transformation is also shift invariant. Hence, there is no need to find $q_0, \dots, q_{n-1}$ separately, and we only need to find a general expression of q_i . Finding q_i is the main technical contribution of this work.

To find such a polynomial q_i , our method works as follows:

Step 1: Find some initial indices (i, j) and a corresponding initial polynomial p_0 in $y_0, \dots, y_{n-1}$ such that

$$\forall j \in [0, n-1] : x_i y_j = p_0 y_j. \quad (3)$$

In other words, we obtain expressions of $x_i y_j$ for some (i, j) . Note that these indices (i, j) may have certain relations, but we do not specify them here as they vary for different SI transformations.

Step 2: Based on these initial relations in Equation 3, recursively find the polynomial p_t in $y_0, \dots, y_{n-1}$ such that

$$\forall j \in [0, n-1], \forall t \geq 0 : x_{i-\lambda t} y_j = p_t y_j.$$

where λ is a constant. Roughly speaking, we obtain expressions of $x_i y_j$ for more (i, j) after this step.

Step 3: According to p_t , find and prove that there exists a polynomial $p_{t'}$ in $y_0, \dots, y_{n-1}$ such that

$$\forall i, j \in [0, n-1] : x_i y_j = p_{t'} y_j.$$

In this way, we collect expressions of $x_i y_j$ for all possible (i, j) .

Step 4: Prove $x_i = p_{t'}$ is the inverse.

In particular, Step 1 is related to the original definition of the SI transformation, and we derive such p_0 directly from its definition, e.g., from $y_i = x_i \oplus \overline{x_{i+1}} x_{i+2}$ for χ_n .

For Step 2, the goal is to find p_t , and this is the key step in our method. For this purpose, we will construct a special ideal and derive recursive relations between p_{t-1} and p_t by studying this ideal. Then, by expanding the recursive relation and using the initial polynomial p_0 , we finally get the complete expression of p_t .

For Step 3, based on the complete expression of p_t , we deduce the target polynomial $p_{t'}$ such that $\forall i, j \in [0, n-1] : x_i y_j = p_{t'} y_j$. For this purpose, we need to show that $i - \lambda t$ will traverse $\{0, \dots, n-1\}$ when t traverses $\{0, \dots, n-1\}$, which naturally gives a condition $\gcd(n, \lambda) = 1$. Moreover, to find such $p_{t'}$, we turn to finding an index t' such that $p_{t'+t}$ remains stable in the sense that $p_{t'+t}$ is essentially the same polynomial as $p_{t'}$ for $\forall t \geq 0$, but with shifted inputs (see Definition 5). This step is also technical and closely related to the formula of the inverse.

At the last step, to show $x_i = p_{t'}$ does hold, we will study two sets and prove that they are identical.

Organization. In Section 2, we establish the foundational elements of the paper, including key definitions and notations that will be used throughout. Then in Section 3, we start showing how to deduce the inverse of χ_n step by step, closely following the above flow. This should help readers understand the main ideas in our approach as χ_n is a very elegant and simple SI transformation. Moreover, at the end of this section, we generalize this method and describe how to use it as well as its limitations. In Section 4 we consider another SI transformation ψ_n used in [8] to demonstrate the versatility of the proposed methodology. Applying the same process as in Section 3, we derive the inverse of ψ_n . Sections 5 and 6 extend the discussion by presenting two additional examples of the inverse of SI transformations, demonstrating that the proposed method is not limited to χ_n or ψ_n , but can be applied to more SI transformations. Finally, we conclude this paper in Section 7 and also give some problems worth further study.

2 Preliminary

In the following, we first give some definitions that are necessary to understand our work. Throughout this paper, we consider Boolean polynomials, i.e., polynomials in $x_0, \dots, x_{n-1}$ defined over the following polynomial quotient ring:

$$\mathbb{F}_2[x_0, \dots, x_{n-1}]/(x_i^2 + x_i : i = 0, \dots, n-1).$$

For convenience, we use $\mathbb{B}[x_0, \dots, x_{n-1}]$ to denote the above polynomial quotient ring, i.e., the set of Boolean polynomials in Boolean variables $x_0, \dots, x_{n-1}$. In other words, $\mathbb{B}[x_0, \dots, x_{n-1}]$ is the following set of polynomials:

$$\left\{ \sum_{i_0, \dots, i_{n-1} \in \{0,1\}} a_{i_0, \dots, i_{n-1}} x_0^{i_0} \cdots x_{n-1}^{i_{n-1}} \mid a_{i_0, \dots, i_{n-1}} \in \mathbb{F}_2 : i_0, \dots, i_{n-1} \in \{0,1\}, x_j^2 + x_j = 0 : 0 \leq j \leq n-1 \right\}.$$

In the following, all definitions are for the ring $\mathbb{B}[x_0, \dots, x_{n-1}]$.

Definition 1. A subset $\mathcal{I} \subseteq \mathbb{B}[x_0, \dots, x_{n-1}]$ is an ideal if it satisfies:

1. $0 \in \mathcal{I}$.
2. If $f, g \in \mathcal{I}$, then $f + g \in \mathcal{I}$.
3. If $f \in \mathcal{I}$ and $h \in \mathbb{B}[x_0, \dots, x_{n-1}]$, $hf \in \mathcal{I}$.

Definition 2. Let $f_0, \dots, f_{m-1}$ be polynomials in $\mathbb{B}[x_0, \dots, x_{n-1}]$. Then, we set

$$\langle f_0, \dots, f_{m-1} \rangle = \left\{ \sum_{i=0}^{m-1} h_i f_i \mid h_0, \dots, h_{m-1} \in \mathbb{B}[x_0, \dots, x_{n-1}] \right\}.$$

Proposition 1. Let $f_0, \dots, f_{m-1}$ be polynomials in $\mathbb{B}[x_0, \dots, x_{n-1}]$. Then, $\langle f_0, \dots, f_{m-1} \rangle$ is an ideal of $\mathbb{B}[x_0, \dots, x_{n-1}]$, and it is called the ideal generated by $f_0, \dots, f_{m-1}$.

Definition 3. Let $f_0, \dots, f_{m-1}$ be polynomials in $\mathbb{B}[x_0, \dots, x_{n-1}]$. Then, we set

$$V(f_0, \dots, f_{m-1}) = \{(a_0, \dots, a_{n-1}) \in \mathbb{F}_2^n \mid f_i(a_0, \dots, a_{n-1}) = 0 \quad \forall i = 0, \dots, m-1\}.$$

We call $V(f_0, \dots, f_{m-1})$ the affine variety defined by $f_0, \dots, f_{m-1}$. Moreover, $|V(f_0, \dots, f_{m-1})|$ denotes the size of the set $V(f_0, \dots, f_{m-1})$.

In other words, an affine variety for polynomials in $\mathbb{B}[x_0, \dots, x_{n-1}]$ is the set of common zeros of a collection of polynomials in an n -dimensional affine space $\mathbb{F}_2^n$.

Proposition 2 (Page 81 [1]). Let $f_0, \dots, f_{m-1}$ be polynomials in $\mathbb{B}[x_0, \dots, x_{n-1}]$. Then,

$$V(f_0, \dots, f_{m-1}) = V(\langle f_0, \dots, f_{m-1} \rangle).$$

For convenience, in the following, we usually write the ideal generated by $f_0, \dots, f_{m-1}$ as

$$\mathcal{F} = \langle f_0, \dots, f_{m-1} \rangle.$$

And according to Proposition 2, we can write the affine variety defined by $f_0, \dots, f_{m-1}$ simply by $V(\mathcal{F})$.

Notation. Before proceeding with a detailed discussion on how to find the inverse of χ_n , we also introduce the following notation that will be used throughout the paper.

1. $\mathbb{Y}$ denotes the polynomial quotient ring $\mathbb{B}[y_0, \dots, y_{n-1}]$.
2. $\mathbb{X}$ denotes the polynomial quotient ring $\mathbb{B}[x_0, \dots, x_{n-1}]$.
3. $\mathbb{K}$ denotes the polynomial quotient ring $\mathbb{B}[x_0, \dots, x_{n-1}, y_0, \dots, y_{n-1}]$.
4. $V(\mathcal{I})$ denotes the affine variety of the ideal $\mathcal{I}$.
5. $|V(\mathcal{I})|$ denotes the number of elements in $V(\mathcal{I})$.
6. (x, y) denotes $(x_0, \dots, x_{n-1}, y_0, \dots, y_{n-1})$.
7. $1^n = (\underbrace{1, 1, \dots, 1}_n)$, $0^n = (\underbrace{0, 0, \dots, 0}_n)$.
8. $[i_0, i_1] = \{i \in \mathbb{N} \mid i_0 \leq i \leq i_1\}$.
9. $a \mid b$ denotes that a divides b , i.e., a is a factor of b , where (a, b) can be integers or polynomials.

Shift invariant transformations. Since we work on SI transformations, we also first give the definition of SI transformations on binary arrays as follows.

Definition 4 (page 96 [2]). A shift $\mathcal{S}_r : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is a transformation that moves each element of an n -bit vector by r positions by right.

$$\mathcal{S}_r(a) = b \iff b_{(i+r) \bmod n} = a_i, \forall i \in [0, n-1].$$

A transformation $\Phi : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ is shift invariant (SI) if

$$\forall a \in \mathbb{F}_2^n, r \in [0, n-1] : \Phi(\mathcal{S}_r(a)) = \mathcal{S}_r(\Phi(a)).$$

Indices of used symbols. In the following sections, to deduce the inverse of SI transformations, many symbols will be used. Here, we emphasize that, throughout this paper, all indices of x_i, y_i, f_i, g_i are within the modulo n . However, for the indices of p_i , they can be any integer. Next, we proceed with detailed analysis of the inverse of χ_n .

3 Finding the Inverse of χ_n

First, we interpret the definition of χ_n from the perspective of solutions to a polynomial system. For this purpose, we introduce the ideal $\mathcal{G} = \langle g_0, \dots, g_{n-1} \rangle$ generated by n polynomials $g_0, \dots, g_{n-1} \in \mathbb{K}$, where

$$\forall i \in [0, n-1] : g_i = y_i \oplus x_i \oplus \overline{x_{i+1}}x_{i+2}. \quad (4)$$

By definition of the affine variety, we have:

$$V(\mathcal{G}) = \{(x, y) \mid g_0 = 0, \dots, g_{n-1} = 0\}.$$

In other words, $V(\mathcal{G})$ is indeed the full mapping table of χ_n , i.e., the set of all input-output pairs (x, y) of χ_n is exactly $V(\mathcal{G})$.

From Equation 1 we can deduce a new equation as follows:

$$\forall (x, y) \in V(\mathcal{G}), \forall i \in [0, n-1] : 0 = x_i \oplus y_i \oplus \overline{x_{i+1}}x_{i+2}.$$

Substituting recursively the value of $x_{i+1} = y_{i+1} \oplus \overline{x_{i+2}}x_{i+3}$. We have,

$$\begin{aligned} 0 &= x_i \oplus y_i \oplus (\overline{y_{i+1}} \oplus \overline{x_{i+2}}x_{i+3})x_{i+2}. \\ 0 &= x_i \oplus y_i \oplus \overline{y_{i+1}}x_{i+2}. \end{aligned}$$

For another ideal $\mathcal{F} = \langle f_0, \dots, f_{n-1} \rangle$ where

$$f_i = x_i \oplus y_i \oplus \overline{x_{i+2}}y_{i+1}, \quad i \in [0, n-1], \quad (5)$$

it has been proved in [10] that $V(\mathcal{G}) = V(\mathcal{F}) \setminus \{(1^n, 0^n)\}$. Indeed, we find that only $V(\mathcal{G}) \subseteq V(\mathcal{F})$ is required for our proof, and we cite the proof in [10], as shown below.

Proof. First, we have

$$\forall (x, y) \in V(\mathcal{G}), i \in [0, n-1] : y_i = x_i \oplus \overline{x_{i+1}}x_{i+2}, y_{i+1} = x_{i+1} \oplus \overline{x_{i+2}}x_{i+3}.$$

Hence,

$$\forall (x, y) \in V(\mathcal{G}), i \in [0, n-1] : x_i \oplus y_i \oplus \overline{y_{i+1}}x_{i+2} = \overline{x_{i+1}}x_{i+2} \oplus \overline{x_{i+1}}x_{i+2} = 0.$$

In other words, $V(\mathcal{G}) \subseteq V(\mathcal{F})$. $\square$

Suppose we do not know in advance that χ_n is invertible when n is an odd number. Indeed, our procedure to find the inverse of χ_n only assumes that n is an odd number but never uses any property of its invertibility. What we want to emphasize here is that our procedure can automatically prove its invertibility when n is an odd number and meanwhile gives the formula of its inverse.

Lemma 1 *Let $f_i = x_i \oplus y_i \oplus x_{i+2}\overline{y_{i+1}}$ for $i \in [0, n-1]$, and $\mathcal{F} = \langle f_0, \dots, f_{n-1} \rangle$. For any given $i, j \in [0, n-1]$, if there exist n polynomials $r_0, \dots, r_{n-1} \in \mathbb{K}$ and one polynomial $r \in \mathbb{K}$ such that*

$$x_i y_j = r \oplus \sum_{k=0}^{n-1} r_k f_k, \quad (6)$$

we can construct $n+1$ polynomials $r'_0, \dots, r'_{n-1}, r' \in \mathbb{K}$ such that

$$x_{i-2} y_j = r' \oplus \sum_{k=0}^{n-1} r'_k f_k, \quad r' = \overline{y_{i-1}}r \oplus y_{i-2} y_j.$$

In particular, we have $r' \oplus x_{i-2} y_j \in \mathcal{F}$.

Remark 1. As already stated in our technical overview, we need expressions of $x_i y_j$ for as many (i, j) as possible. Lemma 1 shows this feasibility, i.e., once we have $x_i y_j \oplus r \in \mathcal{F}$, we can have $x_{i-2} y_j \oplus r' \in \mathcal{F}$, and thus we can generate as many such polynomials as we need. This is also the rationale behind the key construction of f_i and the ideal $\mathcal{F}$ generated by $f_0, \dots, f_{n-1}$.

Note that here we focus on deriving the expression of $x_{i-2} y_j$, as the expression for $x_{i-\lambda} y_j$ involves $x_i y_j$ only when $\lambda = 2$. In this paper, we use the red color to highlight the changes introduced at each step of the proof. This helps make the process easier to follow and understand.

Proof. By definition, we have $f_{i-2} = x_{i-2} \oplus x_i \overline{y_{i-1}} \oplus y_{i-2}$. If there exist such (i, j) and $r_0, \dots, r_{n-1}, r \in \mathbb{K}$ satisfying Equation 6, there will be

$$\begin{aligned} x_{i-2} y_j &= y_j f_{i-2} \oplus \textcolor{red}{x_i y_j} \overline{y_{i-1}} \oplus y_{i-2} y_j, \\ &= y_j f_{i-2} \oplus \overline{y_{i-1}} (\textcolor{red}{r} \oplus \sum_{k=0}^{n-1} \textcolor{red}{r_k} \textcolor{red}{f_k}) \oplus y_{i-2} y_j, \\ &= \underbrace{\overline{y_{i-1}}r \oplus y_{i-2} y_j}_{r'} \oplus \underbrace{(y_j \oplus \overline{y_{i-1}}r_{i-2})}_{r'_{i-2}} f_{i-2} \oplus \sum_{k=0, k \neq i-2}^{n-1} \underbrace{\overline{y_{i-1}}r_k}_{r'_k} f_k \\ &= \overline{y_{i-1}}r \oplus y_{i-2} y_j \oplus \sum_{k=0}^{n-1} r'_k f_k. \end{aligned}$$

This proves the existence of such $r'_0, \dots, r'_{n-1}, r \in \mathbb{K}$, and how to construct them. In particular, $r' = \overline{y_{i-1}}r \oplus y_{i-2} y_j$, and $x_{i-2} y_j \oplus r' = \sum_{k=0}^{n-1} r'_k f_k$. Hence, $r' \oplus x_{i-2} y_j \in \mathcal{F}$. $\square$

In the following theorem, we derive a recursive description of p_t , which can be obtained by analyzing the relationship between p_t and p_{t+1} according to the recursive relation stated Lemma 1, i.e., if $x_i y_j \oplus r \in \mathcal{F}$, $x_{i-2} y_j \oplus r' \in \mathcal{F}$ where $r' = \overline{y_{i-1}}r \oplus y_{i-2} y_j$.

Theorem 1 *Let $f_i = x_i \oplus y_i \oplus x_{i+2}\overline{y_{i+1}}$ for $i \in [0, n-1]$, and $\mathcal{F} = \langle f_0, \dots, f_{n-1} \rangle$. If $\exists i, j \in [0, n-1]$ and $\exists r \in \mathbb{Y}$ such that $y_j \mid r$ and $x_i y_j \oplus r \in \mathcal{F}$, for any p_t satisfying*

$$\begin{cases} p_0 = \frac{r}{y_j}, \\ p_{t+1} = \overline{y_{i-2t-1}}p_t \oplus y_{i-2(t+1)}, \quad t \geq 0, \end{cases} \quad (7)$$

there will be

$$\forall t \geq 0 : (x_{i-2t} \oplus p_t) y_j \in \mathcal{F}, \quad p_t \in \mathbb{Y}$$

Proof. Since $y_j \mid r$ and $r \in \mathbb{Y}$, we have $p_0 = \frac{r}{y_j} \in \mathbb{Y}$. Moreover, since $x_i y_j \oplus r \in \mathcal{F}$ and $r = p_0 y_j$, we have $(x_i \oplus p_0) y_j \in \mathcal{F}$. In this way, we proved the claim for $t = 0$.

By induction, assume it also holds for the case $t = b \geq 0$, i.e.,

$$(x_{i-2b} \oplus p_b) y_j = x_{i-2b} y_j \oplus p_b y_j \in \mathcal{F}, \quad p_b \in \mathbb{Y}.$$

In Lemma 1, it is stated that if $x_u y_v \oplus p \in \mathcal{F}$ for a $p \in \mathbb{Y}$ and a certain pair of indices (u, v) , then $x_{u-2} y_v \oplus \overline{y_{u-1}} p + y_{u-2} y_v \in \mathcal{F}$. In the assumption by induction, we have found such $(u, v) = (i - 2b, j)$ and $p = p_b y_j$. Hence, according to Lemma 1, we have

$$\overline{y_{i-2b-1}} p_b y_j \oplus y_{i-2b-2} y_j \oplus x_{i-2b-2} y_j = (\overline{y_{i-2b-1}} p_b \oplus y_{i-2(b+1)} \oplus x_{i-2(b+1)}) y_j \in \mathcal{F}.$$

Hence, if

$$p_{b+1} = \overline{y_{i-2b-1}} p_b \oplus y_{i-2(b+1)},$$

we have

$$(p_{b+1} \oplus x_{i-2(b+1)}) y_j \in \mathcal{F}.$$

In addition, by the definition of p_{b+1} and as $p_b \in \mathbb{Y}$, we have $p_{b+1} \in \mathbb{Y}$. Therefore, the claim also holds for $t = b + 1$, which completes the proof. $\square$

3.1 Deducing the Inverse of χ_n

For convenience, we denote the inverse of the map χ_n by χ_n^{-1} . To deduce the formula of χ_n^{-1} , we first observe that

$$\forall (x, y) \in V(\mathcal{F}), \forall i \in [0, n-1] : x_i y_{i+1} = f_i y_{i+1} \oplus y_i y_{i+1}, \quad (8)$$

since $f_i = x_i \oplus y_i \oplus x_{i+2} \overline{y_{i+1}}$. Hence, $x_i y_{i+1} \oplus y_i y_{i+1} \in \mathcal{F}$ for $\forall i \in [0, n-1]$. Let

$$\begin{cases} p_0 = y_i, \\ p_{t+1} = \overline{y_{i-2t-1}} p_t \oplus y_{i-2(t+1)}, \forall t \geq 0. \end{cases} \quad (9)$$

According to Theorem 1, we will have

$$\forall t \geq 0 : (x_{i-2t} \oplus p_t) y_{i+1} \in \mathcal{F}, \quad p_t \in \mathbb{Y}.$$

In other words, we have

$$\forall (x, y) \in V(\mathcal{G}) \subseteq V(\mathcal{F}), \forall i \in [0, n-1], \forall t \geq 0 : x_{i-2t} y_{i+1} = p_t y_{i+1}. \quad (10)$$

Indeed, according to the recursive relation specified in Equation 9, we can easily deduce the formula of p_t , which is as follows:

$$\begin{aligned} \forall t \geq 0 : p_t &= \overline{y_{i-2t+1}} p_{t-1} \oplus y_{i-2t} \\ &= \overline{y_{i-2t+1}} (\overline{y_{i-2t+3}} p_{t-2} \oplus y_{i-2(t-1)}) \oplus y_{i-2t} \\ &= \overline{y_{i-2t+1}} \overline{y_{i-2t+3}} p_{t-2} \oplus \overline{y_{i-2t+1}} y_{i-2t+2} \oplus y_{i-2t}. \end{aligned}$$

Substituting recursively, at β th step we have,

$$p_t = \sum_{j=0}^{\beta-1} y_{i-2t+2j+2} \prod_{k=0}^j \overline{y_{i-2t+2k+1}} \oplus \left(\prod_{k=0}^{\beta} \overline{y_{i-2t+2k+1}} \right) p_{t-(\beta+1)} \oplus y_{i-2t}.$$

At $\beta = t - 1$,

$$\begin{aligned} p_t &= \sum_{j=0}^{t-2} y_{i-2t+2j+2} \prod_{k=0}^j \overline{y_{i-2t+2k+1}} \oplus \left(\prod_{k=0}^{t-1} \overline{y_{i-2t+2k+1}} \right) p_0 \oplus y_{i-2t} \\ &= \sum_{j=0}^{t-1} y_{i-2t+2j+2} \prod_{k=0}^j \overline{y_{i-2t+2k+1}} \oplus y_{i-2t}. \end{aligned}$$

By changing the limit of summation from j to $(t - j')$ and the limit of product from k to $(t - k')$ we ultimately obtain:

$$\forall t \geq 0 : p_t = y_{i-2t} \oplus \sum_{j=1}^t y_{i-2j+2} \prod_{k=j}^t \overline{y_{i-2k+1}}. \quad (11)$$

3.2 Finding the Formula

Let us focus on Equation 10, where we have

$$\forall (x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 : x_{i-2t}y_{i+1} = p_t y_{i+1}.$$

Hence, we have

$$\forall (x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 : x_{i-2(\frac{n-1}{2}+t)}y_{i+1} = p_{\frac{n-1}{2}+t}y_{i+1}.$$

Let $\alpha = \frac{n-1}{2}$, i.e., indeed, $2\alpha = -1 \bmod n$, and we have

$$\forall (x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 : x_{i-2t+1}y_{i+1} = p_{\alpha+t}y_{i+1}.$$

In the following, we prove by induction that

$$\forall t \geq 0 : p_{\alpha+t} = y_{i-2t+1} \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2t-2j+2} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2t-2k+1}}. \quad (12)$$

This has been proved for $t = 0$ previously, i.e., setting $t = \frac{n-1}{2}$ in Equation 11. By induction, assume it also holds for $t = b$, and we now prove the case $t = b + 1$. In this case, by the recursive relation in Equation 9, we have

$$\begin{aligned} p_{\alpha+b+1} &= \overline{y_{i-2(\alpha+b)-1}}p_{\alpha+b} \oplus y_{i-2(\alpha+b+1)} \\ &= \overline{y_{i-2b}}p_{\alpha+b} \oplus y_{i-2b-1} \\ &= \overline{y_{i-2b}}(y_{i-2b+1} \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2b-2j+2} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2b-2k+1}}) \oplus y_{i-2b-1} \\ &= y_{i-2b-1} \oplus \overline{y_{i-2b}}y_{i-2b+1} \oplus \sum_{j=2}^{\frac{n-1}{2}} y_{i-2b-2j+2} \prod_{k=j}^{\frac{n-1}{2}+1} \overline{y_{i-2b-2k+1}} \\ &= y_{i-2b-1} \oplus \sum_{j=2}^{\frac{n-1}{2}+1} y_{i-2b-2j+2} \prod_{k=j}^{\frac{n-1}{2}+1} \overline{y_{i-2b-2k+1}} \\ &= y_{i-2b-1} \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2(b+1)-2j+2} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2(b+1)-2k+1}}, \end{aligned}$$

which completes the proof.

The above proof also indicates that

$$\forall t \geq 0 : p_{\alpha+t} = p_{\alpha}(y_{i-2t}, \dots, y_{n-1+i-2t})$$

where

$$p_{\alpha}(y_i, \dots, y_{n-1+i}) = y_{i+1} \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2j+2} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k+1}}.$$

Here, we say p_t becomes **stable** at $t = \alpha$ in the sense that p_t for $t \geq \alpha$ is essentially the same polynomial as p_{α} in $y_i, \dots, y_{i-n+1}$ with a shifted input. A formal definition is as follows.

Definition 5. Let $p_0(y_i, \dots, y_{n-1+i}), \dots, p_{t'}(y_i, \dots, y_{n-1+i}), \dots$ be an infinite set of polynomials in $y_0, \dots, y_{n-1}$. We say p_t becomes stable at $t = t'$ if $\forall t \geq t'$, there exists an integer s such that

$$\forall t \geq t' : p_t(y_i, \dots, y_{n-1+i}) = p_{t'}(y_{i-s}, \dots, y_{n-1+i-s}).$$

According to Equation 12, we have

$$\forall (x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 :$$

$$x_{i-2(\alpha+t)}y_{i+1} = x_{i-2t+1}y_{i+1} = (y_{i-2t+1} \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2t-2j+2} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2t-2k+1}})y_{i+1},$$

When t traverses the set $\{0, \dots, n-1\}$, $1-2t \bmod n$ also traverses the set $\{0, \dots, n-1\}$ since $\gcd(2, n) = 1$ when n is an odd number. Therefore, this is equivalent to

$$\forall (x, y) \in V(\mathcal{G}), \forall i, j \in [0, n-1] : x_i y_j = (y_i \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2j+1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k}})y_j. \quad (13)$$

Indeed, we can easily observe from Equation 13 that the inverse of χ_n is

$$\forall i \in [0, n-1] : x_i = y_i \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2j+1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k}}.$$

To prove the above result, we must show that for any non-zero vector x , there exists a vector y such that $y_j \neq 0$ for some $j \in [0, n-1]$ and $(x, y) \in V(\mathcal{G})$. This can be established using the Lemma 2 given below, which guarantees the existence of such a y for every non-zero x . The proof of the Theorem 2, presented in the next section, relies on the Lemma 2 to demonstrate that for any non-zero input x , a corresponding y with a non-zero component exists.

3.3 Proving the Correctness

Lemma 2 When $n > 0$ is an odd number, $x = 0^n$ is the unique solution of the following polynomial system:

$$\forall i \in [0, n-1] : 0 = x_i \oplus \overline{x_{i+1}}x_{i+2}.$$

Proof. If there exists a solution $(x_0, \dots, x_{n-1})$ such that $x_i = 1$, there will be

$$x_{i-1} = 0, x_{i-2} = 1.$$

Since the indices are within modulo n , and n is an odd number, $i-2j \bmod n$ will traverse $\{0, 1, \dots, n-1\}$ when j traverses $\{0, \dots, n-1\}$. Hence, we must reach the contradiction that $x_j = 0$ and $x_j = 1$ for $\forall j \in [0, n-1]$. On the other hand, it is easy to check that $x = 0^n$ is the solution, and hence it is the unique solution. $\square$

Lemma 3. Let $\mathcal{Z} = \langle z_0, \dots, z_{n-1} \rangle$ and $\mathcal{H} = \langle h_0, \dots, h_{n-1} \rangle$ where

$$\forall i \in [0, n-1] : z_i = x_i \oplus Y_i, \quad h_i = y_i \oplus X_i,$$

and $Y_i \in \mathbb{Y}$ and $X_i \in \mathbb{X}$. If $V(\mathcal{Z}) \subseteq V(\mathcal{H})$ or $V(\mathcal{H}) \subseteq V(\mathcal{Z})$, there is $V(\mathcal{Z}) = V(\mathcal{H})$. If a vectorial Boolean function $B_n : \mathbb{F}_2^n \mapsto \mathbb{F}_2^n$ is defined by

$$\forall i \in [0, n-1] : y_i = X_i,$$

its inverse is

$$\forall i \in [0, n-1] : x_i = Y_i.$$

Proof. Since

$$z_i = x_i \oplus Y_i, \quad \forall i \in [0, n-1],$$

we have $|V(\mathcal{Z})| = 2^n$, i.e., for $\forall x \in \mathbb{F}_2^n$, there is a unique y such that $(x, y) \in V(\mathcal{Z})$. Similarly, we have $|V(\mathcal{H})| = 2^n$. If

$$V(\mathcal{Z}) \subseteq V(\mathcal{H}) \text{ or } V(\mathcal{H}) \subseteq V(\mathcal{Z}),$$

we directly have $V(\mathcal{Z}) = V(\mathcal{H})$. Since $V(\mathcal{Z}) = V(\mathcal{H})$, by the definition of the affine variety, the inverse of B_n is also proved. $\square$

Theorem 2 [10] χ_n is invertible when n is an odd number. In addition, the formula of χ_n^{-1} is

$$\forall i \in [0, n-1] : x_i = y_i \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2j+1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k}}. \quad (14)$$

Proof. Let $\mathcal{W} = \langle w_0, \dots, w_{n-1} \rangle$ where

$$\forall i \in [0, n-1] : w_i = x_i \oplus y_i \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2j+1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k}}.$$

According to Lemma 2, in the affine variety $V(\mathcal{G})$, $y = 0^n$ only when $x = 0^n$. Hence, according to Equation 13, for $\forall (x, y) \in V(\mathcal{G}) \setminus \{(0^n, 0^n)\}$, there always exists an index j' such that $y_{j'} = 1$ and

$$\forall (x, y) \in V(\mathcal{G}) \setminus \{(0^n, 0^n)\}, \forall i \in [0, n-1] : x_i = y_i \oplus \sum_{j=1}^{\frac{n-1}{2}} y_{i-2j+1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k}}.$$

On the other hand, it can be easily verified that $(0^n, 0^n) \in V(\mathcal{W})$. In other words,

$$\forall (x, y) \in V(\mathcal{G}) : (x, y) \in V(\mathcal{W}),$$

which means $V(\mathcal{G}) \subseteq V(\mathcal{W})$. By Lemma 3, we have $V(\mathcal{W}) = V(\mathcal{G})$ and that χ_n is invertible. In addition, the formula of χ_n^{-1} is the one specified in Equation 14. $\square$

3.4 Summary

Compared with [10], how to deduce the formula of χ_n^{-1} is detailed, and it is mainly due to a recursive relation specified in Equation 9. In addition, benefiting from Lemma 3, the proof for $V(\mathcal{W}) \subseteq V(\mathcal{G})$ is simplified, while it is quite long in [10].

As already stated in the technical overview of our approach, the ultimate goal of finding the inverse of χ_n is to find a polynomial q_i in $y_0, \dots, y_{n-1}$ such that

$$\forall i, j \in [0, n-1] : x_i y_j = q_i y_j.$$

However, directly finding such a polynomial is difficult, and we reduce it into several steps, as stated below:

1. Finding an initial polynomial p_0 in $y_0, \dots, y_{n-1}$ such that we can construct expressions of the form $x_i y_j = p_0 y_j$ for some (i, j) . However, this cannot mean $x_i = p_0$ as these initial relations only hold for partial indices (i, j) .
2. Consequently, we aim to increase the range of indices (i, j) so that we can find more expressions of the form $x_i y_j = p_t y_j$. This is reduced to finding the recursive relation between p_t and p_{t-1} .
3. Finally, we need to determine whether we have found expressions of the form $x_i y_j = p_{t'} y_j$ for all possible (i, j) . This is related to showing all indices can be traversed.
4. Finally, to show $x_i = p_{t'}$ is the inverse according to $x_i y_j = p_{t'} y_j$, we first need to prove that p_t is stable at $t = t'$. Then, we prove that $x_i y_j = p_{t'} y_j$ does imply $x_i = p_{t'}$ by considering the relations between two affine varieties.

With the above flow as guidance, we can perform similar analysis of the inverse of several SI transformations, as will be detailed in the following sections. Before this, we first generalize Lemma 1 and Theorem 1, which play a critical role in our method to efficiently find the inverse for more SI transformations.

3.5 Generalizing Lemma 1 and Theorem 1

Lemma 4 *Let λ be an integer constant and $f'_i = f'(y_i, \dots, y_{i+n-1}) \in \mathbb{Y}$ be n Boolean polynomials for $0 \leq i \leq n-1$. Let*

$$f_i = x_i \oplus y_i \oplus (x_{i+\lambda} \oplus c) f'_i \tag{15}$$

be n Boolean polynomials for $i \in [0, n-1]$ where c is a Boolean constant. Let $\mathcal{F} = \langle f_0, \dots, f_{n-1} \rangle$. For any given $i, j \in [0, n-1]$, if there exists a polynomial $r \in \mathbb{K}$ such that

$$x_i y_j \oplus r \in \mathcal{F}, \tag{16}$$

there exists a polynomial $r' \in \mathbb{K}$ such that

$$r' = f'_{i-\lambda} r \oplus y_{i-\lambda} y_j, \quad x_{i-\lambda} y_j \oplus r' = x_{i-\lambda} y_j \oplus f'_{i-\lambda} r \oplus y_{i-\lambda} y_j \in \mathcal{F}. \tag{17}$$

Proof. The proof is almost the same as that for Lemma 1. By definition, we have

$$f_{i-\lambda} = x_{i-\lambda} \oplus (x_i \oplus c)f'_{i-\lambda} \oplus y_{i-\lambda} = x_{i-\lambda} \oplus x_i f'_{i-\lambda} \oplus c f'_{i-\lambda} \oplus y_{i-\lambda}.$$

If there exist such (i, j) and $r \in \mathbb{K}$ satisfying $x_i y_j \oplus r \in \mathcal{F}$, by definition of the ideal $\mathcal{F}$, there must exist n polynomials $r_0, \dots, r_{n-1} \in \mathbb{K}$ such that

$$x_i y_j \oplus r = \sum_{k=0}^{n-1} r_k f_k.$$

Hence, we can derive that

$$\begin{aligned} x_{i-\lambda} y_j &= y_j f_{i-\lambda} \oplus \textcolor{red}{x_i y_j} f'_{i-\lambda} \oplus c y_j f'_{i-\lambda} \oplus y_{i-\lambda} y_j, \\ &= y_j f_{i-\lambda} \oplus f'_{i-\lambda} (r \oplus \sum_{k=0}^{n-1} \textcolor{red}{r_k f_k}) \oplus c y_j f'_{i-\lambda} \oplus y_{i-\lambda} y_j, \\ &= f'_{i-\lambda} (r \oplus c y_j) \oplus y_{i-\lambda} y_j \oplus \underbrace{(y_j \oplus f'_{i-\lambda} r_{i-\lambda})}_{r'_{i-\lambda}} f_{i-\lambda} \oplus \sum_{k=0, k \neq i-\lambda}^{n-1} \underbrace{f'_{i-\lambda} r_k}_{r'_k} f_k \\ &= f'_{i-\lambda} (r \oplus c y_j) \oplus y_{i-\lambda} y_j \oplus \sum_{k=0}^{n-1} r'_k f_k. \end{aligned}$$

Therefore, we can let $r' = f'_{i-\lambda} (r \oplus c y_j) \oplus y_{i-\lambda} y_j$ and there will be $x_{i-\lambda} y_j \oplus r' \in \mathcal{F}$. $\square$

Theorem 3 Let λ be an integer constant and $f'_i = f'(y_i, \dots, y_{i+n-1}) \in \mathbb{Y}$ be n Boolean polynomials for $0 \leq i \leq n-1$. Let

$$f_i = x_i \oplus y_i \oplus (x_{i+\lambda} \oplus c) f'_i \quad (18)$$

be n Boolean polynomials for $i \in [0, n-1]$ where c is a Boolean constant. Let $\mathcal{F} = \langle f_0, \dots, f_{n-1} \rangle$. If $\exists i, j \in [0, n-1]$ and $\exists r \in \mathbb{Y}$ such that r has the factor y_j and $x_i y_j \oplus r \in \mathcal{F}$, for any p_t satisfying

$$\begin{cases} p_0 = \frac{r}{y_j}, \\ p_{t+1} = f'_{i-\lambda(t+1)}(p_t \oplus c) \oplus y_{i-\lambda(t+1)}, \quad t \geq 0, \end{cases} \quad (19)$$

there will be

$$\forall t \geq 0: (x_{i-\lambda t} \oplus p_t) y_j \in \mathcal{F}, \quad p_t \in \mathbb{Y}.$$

Proof. Since the polynomial $r \in \mathbb{Y}$ has the factor y_j , we have $p_0 = \frac{r}{y_j} \in \mathbb{Y}$. Moreover, since $x_i y_j \oplus r \in \mathcal{F}$ and $r = p_0 y_j$, we have $(x_i \oplus p_0) y_j \in \mathcal{F}$. In this way, we proved the claim for $t = 0$. By induction, assume it also holds for the case $t = b \geq 0$, i.e.,

$$(x_{i-\lambda b} \oplus p_b) y_j = x_{i-\lambda b} y_j \oplus p_b y_j \in \mathcal{F}, \quad p_b \in \mathbb{Y}. \quad (20)$$

According to Lemma 4, i.e., replacing the indices (i, j) with $(i - \lambda b, j)$ and the polynomial r with $p_b y_j$ in Equation 16, the following relation can be obtained:

$$r' = f'_{i-\lambda b-\lambda}(p_b y_j \oplus c y_j) \oplus y_{i-\lambda b-\lambda} y_j, \quad x_{i-\lambda b-\lambda} y_j \oplus r' = x_{i-\lambda b-\lambda} y_j \oplus f'_{i-\lambda b-\lambda}(p_b \oplus c) y_j \oplus y_{i-\lambda b-\lambda} y_j \in \mathcal{F}.$$

That is,

$$p_{b+1} = f'_{i-\lambda(b+1)}(p_b \oplus c) \oplus y_{i-\lambda(b+1)} \in \mathbb{Y}, \quad (x_{i-\lambda(b+1)} \oplus p_{b+1}) y_j \in \mathcal{F}$$

$\square$

Application of the general description to χ_n . With the above general description of the lemma and theorem, we can view Lemma 1 and Theorem 1 as concrete instances of the general case by setting

$$\lambda = 2, c = 0, f'_i = \overline{y_{i+1}}, f'(y_0, \dots, y_{n-1}) = \overline{y_1}.$$

For the application to χ_n , apart from the above setting, we also need to set

$$(i, j) = (i, i+1), r = y_i y_{i+1}.$$

Hence, we can derive $x_{i-2t}y_{i+1} = p_t y_{i+1}$ for $i \in [0, n-1]$ and $t \geq 0$ where

$$\begin{cases} p_0 = \frac{r}{y_j} = y_i, \\ p_{t+1} = f'_{i-\lambda(t+1)} p_t \oplus y_{i-\lambda(t+1)} = \overline{y_{i-2t-1}} p_t \oplus y_{i-2(t+1)}, t \geq 0. \end{cases}$$

It will be much clearer that such a general description of the lemma and theorem can be very useful to find the inverse of some SI transformations in the following sections.

How to use our method and what is its limitation? As can be observed from Lemma 4 and Theorem 3, there are some conditions to use them, as listed below:

1. The existence of n such polynomials $f'_0, \dots, f'_{n-1}$.
2. The existence of λ and c .
3. The existence of a polynomial $r \in \mathbb{Y}$ in order to find the inverse.
4. The existence of indices (i, j) .

Hence, finding the inverse with our method is always reduced to solving the above four problems. This may also directly limit the application of our method, as we cannot prove that these problems are always easier to solve than directly finding the inverse. However, we believe this method is still insightful and as it indicates that if all the above four problems can be solved, it is highly possible to find the inverse according to the recursive relation between p_t and p_{t+1} , though we do not have a general proof for this as we have done for Lemma 4 and Theorem 3. Another possible way to extend our method is to further generalize the lemma and theorem, e.g., if f_i is not necessarily of the form $f_i = x_i \oplus y_i \oplus (x_{i+\lambda} \oplus c)f'(y_i, \dots, y_{i+n-1})$.

4 Finding the Inverse of ψ_n

In this section, we study another SI transformation denoted by $\psi_n : \mathbb{F}_2^n \mapsto \mathbb{F}_2^n$ used in practice. Abusing notation, its n -bit input-output pair is still denoted by (x, y) . Moreover, for simplicity, we still use $f_i, g_i, w_i, p_t, \mathcal{G}, \mathcal{F}, \mathcal{W}$ when studying the inverse of ψ_n , and all of them will be redefined here.

The SI transformation ψ_n is defined as follows:

$$\forall i \in [0, n-1] : y_i = x_i \oplus \overline{x_{i+1}} x_{i+2} x_{i+3}.$$

First, we define the ideal $\mathcal{G} = \langle g_0, \dots, g_{n-1} \rangle$ where

$$\forall i \in [0, n-1] : g_i = y_i \oplus x_i \oplus \overline{x_{i+1}} x_{i+2} x_{i+3}.$$

In this way, we have

$$\begin{aligned} \forall (x, y) \in V(\mathcal{G}) : 0 &= y_i \oplus x_i \oplus (\overline{y_{i+1}} \oplus \overline{x_{i+2}} x_{i+3} x_{i+4}) x_{i+2} x_{i+3} \\ &= y_i \oplus x_i \oplus \overline{y_{i+1}} x_{i+2} x_{i+3}, \\ &= y_i \oplus x_i \oplus \overline{y_{i+1}} (y_{i+2} \oplus \overline{x_{i+3}} x_{i+4} x_{i+5}) x_{i+3} \\ &= y_i \oplus x_i \oplus \overline{y_{i+1}} y_{i+2} x_{i+3}. \end{aligned}$$

Let us then define the ideal $\mathcal{F} = \langle f_0, \dots, f_{n-1} \rangle$ where

$$\forall i \in [0, n-1] : f_i = x_i \oplus y_i \oplus \overline{y_{i+1}} y_{i+2} x_{i+3}.$$

According to the above deduction, it holds that $V(\mathcal{G}) \subseteq V(\mathcal{F})$.

Let $\mathcal{F} = \langle f_0, \dots, f_{n-1} \rangle$. Since

$$\forall i \in [0, n-1] : f_i = x_i \oplus y_i \oplus \overline{y_{i+1}} y_{i+2} x_{i+3},$$

we have

$$x_i y_{i+1} \oplus y_i y_{i+1} = f_i y_{i+1} \in \mathcal{F}.$$

Application of Lemma 4 and Theorem 3 to ψ_n . Similar for their application to ψ_n , according to the above analysis, in the application to ψ_n , we can set

$$\begin{aligned}\lambda &= 3, c = 0, \\ f'_i &= \overline{y_{i+1}}y_{i+2}, \quad f'(y_0, \dots, y_{n-1}) = \overline{y_1}y_2, \\ (i, j) &= (i, i+1), \\ r &= y_i y_{i+1}.\end{aligned}$$

Hence, we have the following conclusion for the case of ψ_n :

$$\forall i \in [0, n-1], \forall t \geq 0 : (x_{i-3t} \oplus p_t)y_{i+1} \in \mathcal{F}, \quad (21)$$

where

$$\begin{cases} p_0 = \frac{r}{y_j} = y_i, \\ p_{t+1} = f'_{i-\lambda(t+1)} p_t \oplus y_{i-\lambda(t+1)} = \overline{y_{i-3t-2}} \overline{y_{i-3t-1}} p_t \oplus y_{i-3(t+1)}, \quad t \geq 0. \end{cases} \quad (22)$$

Exploiting the recursive relation in Equation 22, we can easily prove that

$$\forall t \geq 0 : p_t = y_{i-3t} \oplus \sum_{j=1}^t y_{i-3j+3} y_{i-3j+2} \overline{y_{i-3j+1}} \prod_{k=j}^{t-1} y_{i-3k-1} \prod_{k=j}^{t-1} \overline{y_{i-3k-2}}. \quad (23)$$

Proof. From Equation 22

$$\begin{aligned}\forall t \geq 0 : p_t &= \overline{y_{i-3t+1}} y_{i-3t+2} p_{t-1} \oplus y_{i-3t} \\ &= \overline{y_{i-3t+1}} y_{i-3t+2} (\overline{y_{i-3t+4}} y_{i-3t+5} p_{t-2} \oplus y_{i-3(t-1)}) \oplus y_{i-3t} \\ &= \overline{y_{i-3t+1}} y_{i-3t+2} \overline{y_{i-3t+4}} y_{i-3t+5} p_{t-2} \oplus \overline{y_{i-3t+1}} y_{i-3t+2} y_{i-3(t-1)} \oplus y_{i-3t}\end{aligned}$$

Substituting recursively, at β th step we have,

$$\begin{aligned}p_t &= \sum_{j=1}^{\beta} y_{i-3t+3j} y_{i-3t+3j-1} \overline{y_{i-3t+3j-2}} \prod_{k=1}^{j-1} y_{i-3t+3k-1} \prod_{k=1}^{j-1} \overline{y_{i-3t+3k-2}} \\ &\quad \oplus y_{i-3t+3\beta+2} \overline{y_{i-3t+3\beta+1}} \left(\prod_{k=1}^{\beta+1} y_{i-3t+3k-1} \prod_{k=1}^{\beta+1} \overline{y_{i-3t+3k-2}} \right) p_{t-(\beta+1)} \oplus y_{i-3t}\end{aligned}$$

At $\beta = t-1$:

$$\begin{aligned}p_t &= \sum_{j=1}^{t-1} y_{i-3t+3j} y_{i-3t+3j-1} \overline{y_{i-3t+3j-2}} \prod_{k=1}^{j-1} y_{i-3t+3k-1} \prod_{k=1}^{j-1} \overline{y_{i-3t+3k-2}} \\ &\quad \oplus y_{i-1} \overline{y_{i-2}} \left(\prod_{k=1}^t y_{i-3t+3k-1} \prod_{k=1}^t \overline{y_{i-3t+3k-2}} \right) p_0 \oplus y_{i-3t} \\ &= \sum_{j=1}^{t-1} y_{i-3t+3j} y_{i-3t+3j-1} \overline{y_{i-3t+3j-2}} \prod_{k=1}^{j-1} y_{i-3t+3k-1} \prod_{k=1}^{j-1} \overline{y_{i-3t+3k-2}} \\ &\quad \oplus \left(\prod_{k=1}^{t-1} y_{i-3t+3k-1} \prod_{k=1}^{t-1} \overline{y_{i-3t+3k-2}} \right) y_i y_{i-1} \overline{y_{i-2}} \oplus y_{i-3t} \\ &= \sum_{j=1}^t y_{i-3t+3j} y_{i-3t+3j-1} \overline{y_{i-3t+3j-2}} \prod_{k=1}^{j-1} y_{i-3t+3k-1} \prod_{k=1}^{j-1} \overline{y_{i-3t+3k-2}} \oplus y_{i-3t}\end{aligned}$$

By changing the limit of summation from j to $(t-j'+1)$ and the limit of product from k to $(t-k')$ we ultimately obtain Equation 23. $\square$

For example, we have

$$p_1 = y_{i-3} \oplus y_i y_{i-1} \overline{y_{i-2}},$$

$$\begin{aligned}
p_2 &= y_{i-6} \oplus y_i y_{i-1} \overline{y_{i-2} y_{i-4} \overline{y_{i-5}}} \oplus y_{i-3} y_{i-4} \overline{y_{i-5}}, \\
p_3 &= y_{i-9} \oplus y_i y_{i-1} \overline{y_{i-2} y_{i-4} y_{i-7} \overline{y_{i-5} \overline{y_{i-8}}}} \oplus y_{i-3} y_{i-4} \overline{y_{i-5} y_{i-7} \overline{y_{i-8}}} \oplus y_{i-6} y_{i-7} \overline{y_{i-8}}.
\end{aligned}$$

Since $V(\mathcal{G}) \subseteq V(\mathcal{F})$, according to Equation 21, we have

$$\forall(x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 : (x_{i-3t} \oplus p_t) y_{i+1} = 0.$$

In the following, we prove that when $\gcd(n, 3) = 1$, ψ_n is invertible, and give the formula of ψ_n^{-1} based on Equation 23. Since $\gcd(n, 3) = 1$, the inverse of 3 exists in $\mathbb{Z}_n$, and we denote it by θ , i.e.,

$$3\theta = 1 \pmod{n}. \quad (24)$$

Obviously, for $n \geq 4$, we have $\theta \geq 2$. Note that in the following, we only consider the case $n \geq 4$.

Note that in this section we use θ , which satisfies $3\theta = 1 \pmod{n}$. This is different from α used in Section 3.2. The key reason for this distinction is that n can take the form $3k+1$ or $3k+2$ for some $k \geq 0$, such that $\gcd(n, 3) = 1$. In contrast to considering two separate cases for α , namely $\alpha = \frac{3n-1}{3}$ and $\alpha = \frac{3n-2}{3}$, we introduce θ as a unified solution that works for both forms of α .

4.1 Finding the Formula

According to Equation 21, we have

$$\forall(x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 : x_{i-3(\theta+t)} y_{i+1} = p_{\theta+t} y_{i+1}. \quad (25)$$

By induction, we can prove

$$\forall t \geq 0 : p_{\theta+t} = y_{i-3t-1} \oplus \sum_{j=t+1}^{\theta-1+t} y_{i-3j} y_{i-3j-1} \overline{y_{i-3j-2}} \prod_{k=j}^{\theta-2+t} y_{i-3k-4} \prod_{k=j}^{\theta-2+t} \overline{y_{i-3k-5}}. \quad (26)$$

First, the case $t = 0$ has been proved, i.e., refer to Equation 23. Assume that it also holds for the case $t = b$, and we now prove the case $t = b+1$. By the recursive relation in Equation 22, we have

$$\begin{aligned}
p_{\theta+b+1} &= \overline{y_{i-3(\theta+b)-2} y_{i-3(\theta+b)-1} p_{\theta+b}} \oplus y_{i-3(\theta+b+1)} \\
&= \overline{y_{i-3b-3} y_{i-3b-2} p_{\theta+b}} \oplus y_{i-3b-4} \\
&= y_{i-3b-4} \oplus \overline{y_{i-3b-3} y_{i-3b-2} y_{i-3b-1}} \\
&\quad \oplus \overline{y_{i-3b-3} y_{i-3b-2}} \left(\sum_{j=b+1}^{\theta-1+b} y_{i-3j} y_{i-3j-1} \overline{y_{i-3j-2}} \prod_{k=j}^{\theta-2+b} y_{i-3k-4} \prod_{k=j}^{\theta-2+b} \overline{y_{i-3k-5}} \right) \\
&= y_{i-3b-4} \oplus \overline{y_{i-3b-3} y_{i-3b-2} y_{i-3b-1}} \\
&\quad \oplus \overline{y_{i-3b-3} y_{i-3b-2}} \left(\sum_{j=b+2}^{\theta-1+b} y_{i-3j} y_{i-3j-1} \overline{y_{i-3j-2}} \prod_{k=j}^{\theta-2+b} y_{i-3k-4} \prod_{k=j}^{\theta-2+b} \overline{y_{i-3k-5}} \right) \\
&= y_{i-3b-4} \oplus \overline{y_{i-3b-3} y_{i-3b-2} y_{i-3b-1}} \\
&\quad \oplus \sum_{j=b+2}^{\theta-1+b} y_{i-3j} y_{i-3j-1} \overline{y_{i-3j-2}} \prod_{k=j}^{\theta-1+b} y_{i-3k-4} \prod_{k=j}^{\theta-1+b} \overline{y_{i-3k-5}} \\
&= y_{i-3b-4} \oplus \sum_{j=b+2}^{\theta+b} y_{i-3j} y_{i-3j-1} \overline{y_{i-3j-2}} \prod_{k=j}^{\theta-1+b} y_{i-3k-4} \prod_{k=j}^{\theta-1+b} \overline{y_{i-3k-5}}.
\end{aligned}$$

Hence, the correctness of Equation 4.1 is proved.

Indeed, Equation 4.1 can also be written as

$$\forall t \geq 0 : p_{\theta+t} = y_{i-3t-1} \oplus \sum_{j=1}^{\theta-1} y_{i-3j-3t} y_{i-3j-1} \overline{y_{i-3j-2}} \prod_{k=j}^{\theta-2} y_{i-3t-3k-4} \prod_{k=j}^{\theta-2} \overline{y_{i-3t-3k-5}}.$$

Therefore, according to Equation 25, we have

$$\forall(x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 :$$

$$x_{i-3t-1}y_{i+1} = (y_{i-3t-1} \oplus \sum_{j=1}^{\theta-1} y_{i-3j-3t} \overline{y_{i-3j-1}} \prod_{k=j}^{\theta-2} y_{i-3k-4} \prod_{k=j}^{\theta-2} \overline{y_{i-3k-5}}) y_{i+1}.$$

As $\gcd(n, 3) = 1$, $-3t - 1 \pmod n$ will traverse $\{0, \dots, n-1\}$ when t traverses $\{0, \dots, n-1\}$. Hence, we have

$$\begin{aligned} \forall (x, y) \in V(\mathcal{G}), \forall i, j \in [0, n-1] : \\ x_i y_j = (y_i \oplus \sum_{j=1}^{\theta-1} y_{i-3j+1} y_{i-3j} \overline{y_{i-3j-1}} \prod_{k=j}^{\theta-2} y_{i-3k-3} \prod_{k=j}^{\theta-2} \overline{y_{i-3k-4}}) y_j. \end{aligned} \quad (27)$$

4.2 Proving the Correctness

Lemma 5 *For any $n \geq 4$, $x = 0^n$ is the unique solution of the following polynomial system:*

$$\forall i \in [0, n-1] : 0 = x_i \oplus \overline{x_{i+1}} x_{i+2} x_{i+3}.$$

Proof. If there exists a solution $(x_0, \dots, x_{n-1})$ such that $x_i = 1$, there will be

$$x_{i-1} = 0, x_{i+1} = 0, x_{i+2} = 1, x_{i+3} = 1.$$

From $x_{i+2} = 1$, similarly we can deduce

$$x_{i+1} = 0, x_{i+3} = 0, x_{i+4} = 1, x_{i+5} = 1.$$

Hence, there is a contradiction in the value of x_{i+3} . On the other hand, it is easy to check that $x = 0^n$ is the solution, and hence it is the unique solution. $\square$

Remark 2. One may observe that in our method for χ_n^{-1} , the condition $\gcd(n, 2) = 1$ is not only used to deduce an expression of $x_i y_j$ (i.e., Equation 13) that holds for $\forall i, j \in [0, n-1]$, but also to prove Lemma 2. However, in the case of ψ_n^{-1} , the condition $\gcd(n, 3) = 1$ is only used to deduce a similar expression of $x_i y_j$ (i.e., Equation 27) that holds for all $i, j \in [0, n-1]$, while this is not required in Lemma 5. Therefore, it indicates that such a condition is not directly related to ensuring 0^n is mapped to 0^n in both directions. Instead, it is more related to finding the formula of the inverse, i.e., note that we can observe the formula from the expression of $x_i y_j$. Hence, such a condition is possibly deeply related to the invertibility of SI transformations.

Theorem 4 *ψ_n is invertible when $\gcd(n, 3) = 1$. In addition, the formula of ψ_n^{-1} is*

$$\forall i \in [0, n-1] : x_i = y_i \oplus \sum_{j=1}^{\theta-1} y_{i-3j+1} y_{i-3j} \overline{y_{i-3j-1}} \prod_{k=j}^{\theta-2} y_{i-3k-3} \prod_{k=j}^{\theta-2} \overline{y_{i-3k-4}}, \quad (28)$$

Proof. Let $\mathcal{W} = \langle w_0, \dots, w_{n-1} \rangle$ be the ideal generated by

$$w_i = x_i \oplus y_i \oplus \sum_{j=1}^{\theta-1} y_{i-3j+1} y_{i-3j} \overline{y_{i-3j-1}} \prod_{k=j}^{\theta-2} y_{i-3k-3} \prod_{k=j}^{\theta-2} \overline{y_{i-3k-4}}.$$

By applying Lemma 5 and referring to Equation 27, for $\forall (x, y) \in V(\mathcal{G})$ where $y \neq 0^n$, there must exist an index j' such that $y_{j'} = 1$, and we then have

$$x_i = y_i \oplus \sum_{j=1}^{\theta-1} y_{i-3j+1} y_{i-3j} \overline{y_{i-3j-1}} \prod_{k=j}^{\theta-2} y_{i-3k-3} \prod_{k=j}^{\theta-2} \overline{y_{i-3k-4}}.$$

Moreover, for $(x, y) \in V(\mathcal{G})$ where $y = 0^n$, according to Lemma 5, there must be $x = 0^n$. As $(x, y) = (0^n, 0^n) \in V(\mathcal{W})$, we then have

$$\forall (x, y) \in V(\mathcal{G}) : (x, y) \in V(\mathcal{W}).$$

Therefore, $V(\mathcal{G}) \subseteq V(\mathcal{W})$. According to Lemma 3, $V(\mathcal{G}) = V(\mathcal{W})$ and ψ_n is invertible, and its inverse is given by Equation 28. $\square$

Following the same approach as in the previous section, we now consider two new SI transformations ζ_n and ϕ_n .

5 Finding the Inverse of ζ_n

In this section, we study another SI transformation denoted by $\zeta_n : \mathbb{F}_2^n \mapsto \mathbb{F}_2^n$ that can be used in practice. Abusing notation, its n -bit input-output pair is still denoted by (x, y) . Moreover, for simplicity, we still use $f_i, g_i, w_i, p_t, \mathcal{G}, \mathcal{F}, \mathcal{W}$ when studying the inverse of ζ_n , and all of them will be redefined here.

The SI transformation ζ_n is defined as follows:

$$\forall i \in [0, n-1] : y_i = x_i \oplus \overline{x_{i-1}} \overline{x_{i+1}} x_{i+2}.$$

First, we define the ideal $\mathcal{G} = \langle g_0, \dots, g_{n-1} \rangle$ where

$$\forall i \in [0, n-1] : g_i = y_i \oplus x_i \oplus \overline{x_{i-1}} \overline{x_{i+1}} x_{i+2}.$$

In this way, we have

$$\begin{aligned} \forall (x, y) \in V(\mathcal{G}) : 0 &= y_i \oplus x_i \oplus (\overline{y_{i-1}} \oplus \overline{x_{i-2}} \overline{x_{i+1}}) \overline{x_{i+1}} x_{i+2} \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} \overline{x_{i+1}} x_{i+2}, \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} (\overline{y_{i+1}} \oplus \overline{x_i} \overline{x_{i+2}} x_{i+3}) x_{i+2} \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} x_{i+2}. \end{aligned}$$

Let us then define the ideal $\mathcal{F} = \langle f_0, \dots, f_{n-1} \rangle$ where

$$\forall i \in [0, n-1] : f_i = x_i \oplus y_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} x_{i+2}.$$

According to the above deduction, it holds that $V(\mathcal{G}) \subseteq V(\mathcal{F})$. Since $f_i = x_i \oplus y_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} x_{i+2}$, $x_i y_{i+1} \oplus y_i y_{i+1} \in \mathcal{F}$ for $\forall i \in [0, n-1]$.

Application of Lemma 4 and Theorem 3 to ζ_n . Similarly, in the application to ζ_n , we can set

$$\begin{aligned} \lambda &= 2, c = 0, \\ f'_i &= \overline{y_{i-1}} \overline{y_{i+1}}, f'(y_0, \dots, y_{n-1}) = \overline{y_{n-1}} \overline{y_1}, \\ (i, j) &= (i, i+1), \\ r &= y_i y_{i+1}. \end{aligned}$$

Hence, we have the following conclusion for the case of ζ_n :

$$\forall i \in [0, n-1], \forall t \geq 0 : (x_{i-2t} \oplus p_t) y_{i+1} \in \mathcal{F}, \quad (29)$$

where

$$\begin{cases} p_0 = \frac{r}{y_j} = y_i, \\ p_{t+1} = f'_{i-\lambda(t+1)} p_t \oplus y_{i-\lambda(t+1)} = \overline{y_{i-2t-3}} \overline{y_{i-2t-1}} p_t \oplus y_{i-2(t+1)}, \quad t \geq 0. \end{cases} \quad (30)$$

5.1 Deducing the Inverse of ζ_n

For convenience, we denote the inverse of the map ζ_n by ζ_n^{-1} . According to the recursive relation specified in Equation 30, we can easily deduce the formula of p_t , which is as follows:

$$\forall t \geq 0 : p_t = y_{i-2t} \oplus \sum_{j=0}^{t-1} y_{i-2j} \prod_{k=j}^t \overline{y_{i-2k-1}}. \quad (31)$$

We now omit the intermediate steps for deriving p_t from recursive relation Equation 30. The process can be carried out in a manner similar to the derivations presented before, where the recursive relations were analyzed and solved step by step. By following the same methodology, one can obtain the expression for p_t from its recursive relation, though the detailed steps are not repeated here for brevity.

5.2 Finding the Formula

Let us focus on Equation 29, since $V(\mathcal{G}) \subseteq V(\mathcal{F})$, we have

$$\forall(x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 : x_{i-2t}y_{i+1} = p_t y_{i+1}.$$

Hence, we have

$$\forall(x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 : x_{i-2(\frac{n-1}{2}+t)}y_{i+1} = p_{\frac{n-1}{2}+t}y_{i+1}.$$

Let $\alpha = \frac{n-1}{2}$, i.e., indeed, $2\alpha = -1 \pmod n$, and we have

$$\forall(x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 : x_{i-2t+1}y_{i+1} = p_{\alpha+t}y_{i+1}.$$

In the following, we prove by induction that

$$\forall t \geq 0 : p_{\alpha+t} = y_{i-2t+1} \oplus \sum_{j=1}^{\frac{n-1}{2}-1} y_{i-2t-2j} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2t-2k-1}}.$$

Let us start with $t = 0$, setting $t = \alpha = \frac{n-1}{2}$ in Equation 31, we have,

$$p_{\alpha} = y_{i-2\alpha} \oplus \sum_{j=0}^{\alpha-1} y_{i-2j} \prod_{k=j}^{\alpha} \overline{y_{i-2k-1}}$$

The second term (at $j = 0$) on the left-hand side of the above equation equals $y_i(\overline{y_{i-1}} \cdots \overline{y_{i-2\alpha-1}}) = y_i(\overline{y_{i-1}} \cdots \overline{y_i}) = 0$. Thus the above equation can be reduced to

$$p_{\alpha} = y_{i-2\alpha} \oplus \sum_{j=1}^{\alpha-1} y_{i-2j} \prod_{k=j}^{\alpha} \overline{y_{i-2k-1}}$$

Thus, it holds for $t = 0$. By induction, assume that it also holds for $t = b$, and we now prove the case $t = b + 1$. In this case, by the recursive relation in Equation 30, we have

$$\begin{aligned} p_{\alpha+b+1} &= \overline{y_{i-2(\alpha+b)-3}} \overline{y_{i-2(\alpha+b)-1}} p_{\alpha+b} \oplus y_{i-2(\alpha+b+1)} \\ &= \overline{y_{i-2b-2}} \overline{y_{i-2b}} p_{\alpha+b} \oplus y_{i-2b-1} \\ &= \overline{y_{i-2b-2}} \overline{y_{i-2b}} (y_{i-2b+1} \oplus \sum_{j=1}^{\frac{n-1}{2}-1} y_{i-2b-2j} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2b-2k-1}}) \oplus y_{i-2b-1} \\ &= y_{i-2b-1} \oplus (\overline{y_{i-2b-2}} \overline{y_{i-2b}} y_{i-2b+1} \oplus \sum_{j=2}^{\frac{n-1}{2}-1} y_{i-2b-2j} \prod_{k=j}^{\frac{n-1}{2}+1} \overline{y_{i-2b-2k-1}}) \\ &= y_{i-2b-1} \oplus (\sum_{j=2}^{\frac{n-1}{2}-1} y_{i-2b-2j} \prod_{k=j}^{\frac{n-1}{2}+1} \overline{y_{i-2b-2k-1}}) \oplus \overline{y_{i-2b-2}} \overline{y_{i-2b}} y_{i-2b+1} \\ &= y_{i-2b-1} \oplus \sum_{j=2}^{\frac{n-1}{2}} y_{i-2b-2j} \prod_{k=j}^{\frac{n-1}{2}+1} \overline{y_{i-2b-2k-1}} \\ &= y_{i-2(b+1)+1} \oplus \sum_{j=1}^{\frac{n-1}{2}-1} y_{i-2(b+1)-2j} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2(b+1)-2k-1}} \end{aligned}$$

which completes the proof. By the above proof, we have

$$\begin{aligned} \forall(x, y) \in V(\mathcal{G}), \forall i \in [0, n-1], \forall t \geq 0 : \\ x_{i-2(\alpha+t)}y_{i+1} = x_{i-2t+1}y_{i+1} = (y_{i-2t+1} \oplus \sum_{j=1}^{\frac{n-1}{2}-1} y_{i-2t-2j} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2t-2k-1}})y_{i+1}, \end{aligned}$$

When t traverses the set $\{1, \dots, n\}$, $1 - 2t \bmod n$ traverses the set $\{0, \dots, n-1\}$ since $\gcd(2, n) = 1$ when n is an odd number. Therefore, this is equivalent to

$$\forall (x, y) \in V(\mathcal{G}), \forall i, j \in [0, n-1] : x_i y_j = (y_i \oplus \sum_{j=1}^{\frac{n-1}{2}-1} y_{i-2j-1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k-2}}) y_j. \quad (32)$$

Indeed, we can easily observe from Equation 32 that the inverse of ζ_n is

$$\forall i \in [0, n-1] : x_i = y_i \oplus \sum_{j=1}^{\frac{n-1}{2}-1} y_{i-2j-1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k-2}}.$$

A simple proof is described in the following section.

5.3 Proving the Correctness

Lemma 6 *When $n > 0$ is an odd number, $x = 0^n$ is the unique solution of the following polynomial system:*

$$\forall i \in [0, n-1] : 0 = x_i \oplus \overline{x_{i-1}} \overline{x_{i+1}} x_{i+2}.$$

Proof. If there exists a solution $(x_0, \dots, x_{n-1})$ such that $x_i = 1$, there will be

$$x_{i-1} = 0, x_{i+1} = 0, x_{i+2} = 1.$$

Since the indices are modulo n , and n is an odd number, $i + 2j \bmod n$ will traverse $\{0, 1, \dots, n-1\}$ when j traverses $\{0, \dots, n-1\}$. Hence, we must reach the contradiction that $x_j = 0$ and $x_j = 1$ for $\forall j \in [0, n-1]$. On the other hand, it is easy to check that $x = 0^n$ is the solution, and hence it is the unique solution. $\square$

Theorem 5 ζ_n is invertible when n is an odd number. In addition, the formula of ζ_n^{-1} is

$$\forall i \in [0, n-1] : x_i = y_i \oplus \sum_{j=1}^{\frac{n-1}{2}-1} y_{i-2j-1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k-2}}. \quad (33)$$

Proof. Let $\mathcal{W} = \langle w_0, \dots, w_{n-1} \rangle$ where

$$\forall i \in [0, n-1] : w_i = x_i \oplus y_i \oplus \sum_{j=1}^{\frac{n-1}{2}-1} y_{i-2j-1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k-2}}.$$

Applying Lemma 6, and referencing Equation 32, for $\forall (x, y) \in V(\mathcal{G}) \setminus \{(0^n, 0^n)\}$, there always exists an index j' such that $y_{j'} = 1$ and

$$\forall (x, y) \in V(\mathcal{G}) \setminus \{(0^n, 0^n)\}, \forall i \in [0, n-1] : x_i = y_i \oplus \sum_{j=1}^{\frac{n-1}{2}-1} y_{i-2j-1} \prod_{k=j}^{\frac{n-1}{2}} \overline{y_{i-2k-2}}.$$

On the other hand, it can be easily verified that $(0^n, 0^n) \in V(\mathcal{W})$. In other words,

$$\forall (x, y) \in V(\mathcal{G}) : (x, y) \in V(\mathcal{W}),$$

which means $V(\mathcal{G}) \subseteq V(\mathcal{W})$. By Lemma 3, we have $V(\mathcal{W}) = V(\mathcal{G})$ and that ζ_n is invertible. In addition, the formula of ζ_n^{-1} is the one specified in Equation 33. $\square$

6 Finding the Inverse of ϕ_n

The SI transformation $\phi_n : \mathbb{F}_2^n \mapsto \mathbb{F}_2^n$ is defined as follows:

$$\forall i \in [0, n-1] : y_i = x_i \oplus \overline{x_{i-1}} \overline{x_{i+1}} x_{i+2} x_{i+3}.$$

Theorem 6 ϕ_n is invertible $\forall n \geq 1$. In addition, the formula of ϕ_n^{-1} is

$$\forall i \in [0, n-1] : x_i = y_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} y_{i+2} y_{i+3}. \quad (34)$$

Proof. First, we define the ideal $\mathcal{G} = \langle g_0, \dots, g_{n-1} \rangle$ where

$$\forall i \in [0, n-1] : g_i = y_i \oplus x_i \oplus \overline{x_{i-1}} \overline{x_{i+1}} x_{i+2} x_{i+3}.$$

In this way, we have

$$\begin{aligned} \forall (x, y) \in V(\mathcal{G}) : 0 &= y_i \oplus x_i \oplus (\overline{y_{i-1}} \oplus \overline{x_{i-2}} \overline{x_{i+1}} x_{i+2}) \overline{x_{i+1}} x_{i+2} x_{i+3} \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} \overline{x_{i+1}} x_{i+2} x_{i+3}, \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} (\overline{y_{i+1}} \oplus \overline{x_i} \overline{x_{i+2}} x_{i+3} x_{i+4}) x_{i+2} x_{i+3} \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} x_{i+2} x_{i+3}. \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} (y_{i+2} \oplus \overline{x_{i+1}} \overline{x_{i+3}} x_{i+4} x_{i+5}) x_{i+3}. \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} y_{i+2} x_{i+3}. \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} y_{i+2} (y_{i+3} \oplus \overline{x_{i+2}} \overline{x_{i+4}} x_{i+5} x_{i+6}). \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} y_{i+2} y_{i+3} \oplus \underbrace{\overline{y_{i-1}} \overline{y_{i+1}} y_{i+2} \overline{x_{i+2}} \overline{x_{i+4}} x_{i+5} x_{i+6}}_0 \\ &= y_i \oplus x_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} y_{i+2} y_{i+3} \end{aligned}$$

Let us then define the ideal $\mathcal{W} = \langle w_0, \dots, w_{n-1} \rangle$ where

$$\forall i \in [0, n-1] : w_i = x_i \oplus y_i \oplus \overline{y_{i-1}} \overline{y_{i+1}} y_{i+2} y_{i+3}.$$

According to the above deduction, it holds that $V(\mathcal{G}) \subseteq V(\mathcal{W})$. By Lemma 3, we have $V(\mathcal{W}) = V(\mathcal{G})$ and that ϕ_n is invertible. In addition, the formula of ϕ_n^{-1} is given by Equation 34. It is clear that ϕ_n is self invertible and invertible for all $n \geq 1$ which eliminates the necessity of relying on any lemmas or theorems to establish its inverse. $\square$

7 Conclusion

In this work, we successfully developed a general and systematic method for deriving the inverse of some SI transformations. Our approach simplified the existing proof for the inverse of χ_n in [10], making it easier to follow. In addition, we have introduced a novel formula for the inverse of ψ_n that also has a practical application in cryptography. The methods we have developed are not limited to χ_n and ψ_n ; they are versatile and can be extended to more SI transformations. The ability to derive explicit inverse functions is particularly valuable for designing secure and efficient cryptographic algorithms, where understanding reversibility plays a crucial role. These findings open up exciting opportunities for further research into the algebraic properties of SI transformations and their applications in both symmetric key cryptography and in the design of hash functions.

7.1 Future Direction

There are some important directions for future work that should be explored:

1. How to prove and find the inverse without expanding the recursive description of p_t ? We tried this, but failed. Note that the obstacle is that we cannot know when p_t will become stable, or prove it will be stable at $t = t'$ for a given t' , without expanding the recursive description of p_t . One may observe that the recursive description is very simple, but the expanded description is quite complex. Hence, if this can be solved, we can give a proven elegant recursive description of the inverse.
2. Is $\gcd(n, \lambda) = 1$ a sufficient and necessary condition for invertibility? Can we prove the inverse with only partial expressions of $x_i y_j = p_t y_j$ where (i, j) does not traverse all possible values? If possible, such a condition might be relaxed.
3. How to further develop a more generalized method that can work for any SI transformations? A possible direction might be to further generalize Lemma 4 and Theorem 3, as there are still too many restrictions on the ideal $\mathcal{F}$ generated by $f_0, \dots, f_{n-1}$.
4. Is there a general method to efficiently find the polynomials $f_0, \dots, f_{n-1}$ and r for any concrete SI transformation in order to use Lemma 4 and Theorem 3? Our current method is more like hand-crafted.

No.	Definition	Construction of f_i	construction of r	Construction of f'	λ	c	(i, j)
1	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+2}x_{i+3}}$	$x_i = y_i + \overline{y_{i+1}} \overline{y_{i+2}x_{i+3}}$	$y_i y_{i+1}$	$\overline{y_1} \overline{y_2}$	3	0	$(i, i+1)$
2	$y_i = x_i + x_{i+1} \overline{x_{i+2}} \overline{x_{i+3}}$	$x_i = y_i + y_{i+1} \overline{y_{i+2}} \overline{x_{i+3}}$	$y_i y_{i+2}$	$y_1 \overline{y_2}$	3	1	$(i, i+2)$
3	$y_i = x_i + \overline{x_{i-1}} x_{i+2} \overline{x_{i+3}}$	$x_i = y_i + \overline{y_{i-1}} x_{i+2} \overline{y_{i+3}}$	$y_i y_{i+3}$	$\overline{y_{n-1}} \overline{y_3}$	2	0	$(i, i+3)$
4	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+3}x_{i+4}}$	$x_i = y_i + \overline{y_{i+1}} \overline{y_{i+3}x_{i+4}}$	$y_i y_{i+1}$	$\overline{y_1} \overline{y_3}$	4	0	$(i, i+1)$
5	$y_i = x_i + \overline{x_{i-1}} \overline{x_{i+1}x_{i+2}x_{i+3}}$	$x_i = y_i + \overline{y_{i-1}} \overline{y_{i+1}x_{i+2}y_{i+3}}$	$y_i y_{i+1}$	$\overline{y_{n-1}} \overline{y_1} \overline{y_3}$	2	0	$(i, i+1)$
6	$y_i = x_i + \overline{x_{i-1}} \overline{x_{i+1}} \overline{x_{i+2}x_{i+3}}$	$x_i = y_i + \overline{y_{i-1}} \overline{y_{i+1}} \overline{y_{i+2}x_{i+3}}$	$y_i y_{i+1}$	$\overline{y_{n-1}} \overline{y_1} \overline{y_2}$	3	0	$(i, i+1)$
7	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+2}x_{i+3}x_{i+4}}$	$x_i = y_i + \overline{y_{i+1}} \overline{y_{i+2}y_{i+3}x_{i+4}}$	$y_i y_{i+1}$	$\overline{y_1} \overline{y_2} \overline{y_3}$	4	0	$(i, i+1)$
8	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+2}} \overline{x_{i+3}x_{i+4}}$	$x_i = y_i + \overline{y_{i+1}} \overline{y_{i+2}} \overline{y_{i+3}x_{i+4}}$	$y_i y_{i+1}$	$\overline{y_1} \overline{y_2} \overline{y_3}$	4	0	$(i, i+1)$
9	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+2}x_{i+3}x_{i+4}}$	$x_i = y_i + \overline{y_{i+1}} \overline{y_{i+2}y_{i+3}x_{i+4}}$	$y_i y_{i+1}$	$\overline{y_1} \overline{y_2} \overline{y_3}$	4	0	$(i, i+1)$
10	$y_i = x_i + \overline{x_{i-1}} \overline{x_{i+1}x_{i+2}} \overline{x_{i+4}}$	$x_i = y_i + \overline{y_{i-1}} \overline{y_{i+1}y_{i+2}} \overline{x_{i+4}}$	$y_i y_{i+1}$	$\overline{y_{n-1}} \overline{y_1} \overline{y_2}$	4	1	$(i, i+1)$
11	$y_i = x_i + \overline{x_{i-2}x_{i+2}x_{i+3}x_{i+5}}$	$x_i = y_i + \overline{y_{i-2}y_{i+2}y_{i+3}y_{i+5}}$		Not required			
12	$y_i = x_i + \overline{x_{i-2}} \overline{x_{i-1}x_{i+1}} \overline{x_{i+2}}$	$x_i = y_i + \overline{y_{i-2}} \overline{y_{i-1}y_{i+1}} \overline{y_{i+2}}$		Not required			
13	$y_i = x_i + \overline{x_{i-2}x_{i-1}} \overline{x_{i+1}x_{i+2}}$	$x_i = y_i + \overline{y_{i-2}y_{i-1}} \overline{y_{i+1}y_{i+2}}$		Not required			
14	$x_i = x_i + \overline{x_{i-2}x_{i+1}} \overline{x_{i+2}} \overline{x_{i+3}}$	$x_i = y_i + \overline{y_{i-2}y_{i+1}} \overline{y_{i+2}} \overline{y_{i+3}}$		Not required			

Table 1: More examples for our method to apply, where the last 4 functions are self invertible, and finding their inverse does not require any lemma or theorem to prove their inverse.

No.	Definition	p_t
1	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+2}x_{i+3}}$	$y_{i-3t} + \sum_{j=0}^{t-1} y_{i-3j} \prod_{k=j+1}^t \overline{y_{i-3k+1}} \overline{y_{i-3k+2}}$
2	$y_i = x_i + x_{i+1} \overline{x_{i+2}} \overline{x_{i+3}}$	$y_{i-3t} + \sum_{j=0}^{t-1} y_{i-3j} \prod_{k=j+1}^t y_{i-3k+1} \overline{y_{i-3j+2}} + \prod_{k=1}^t y_{i-3k+1} \overline{y_{i-3j+2}}$
3	$y_i = x_i + \overline{x_{i-1}} x_{i+2} \overline{x_{i+3}}$	$y_{i-2t} + \sum_{j=0}^{t-1} y_{i-2j} \prod_{k=j+1}^t \overline{y_{i-2k-1}} \overline{y_{i-2k+3}}$
4	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+3}x_{i+4}}$	$y_{i-4t} + \sum_{j=0}^{t-1} y_{i-4j} \prod_{k=j+1}^t \overline{y_{i-4k+1}} \overline{y_{i-4k+3}}$
5	$y_i = x_i + \overline{x_{i-1}} \overline{x_{i+1}x_{i+2}x_{i+3}}$	$y_{i-2t} + \sum_{j=0}^{t-1} y_{i-2j} \prod_{k=j+1}^t \overline{y_{i-2k-1}} \overline{y_{i-2k+1}} \overline{y_{i-2k+3}}$
6	$y_i = x_i + \overline{x_{i-1}} \overline{x_{i+1}} \overline{x_{i+2}x_{i+3}}$	$y_{i-3t} + \sum_{j=0}^{t-1} y_{i-3j} \prod_{k=j+1}^t \overline{y_{i-3k-1}} \overline{y_{i-3k+1}} \overline{y_{i-3k+2}}$
7	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+2}x_{i+3}x_{i+4}}$	$y_{i-4t} + \sum_{j=0}^{t-1} y_{i-4j} \prod_{k=j+1}^t \overline{y_{i-4k+1}} \overline{y_{i-4k+2}} \overline{y_{i-4k+3}}$
8	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+2}} \overline{x_{i+3}x_{i+4}}$	$y_{i-4t} + \sum_{j=0}^{t-1} y_{i-4j} \prod_{k=j+1}^t \overline{y_{i-4k+1}} \overline{y_{i-4k+2}} \overline{y_{i-4k+3}}$
9	$y_i = x_i + \overline{x_{i+1}} \overline{x_{i+2}x_{i+3}x_{i+4}}$	$y_{i-4t} + \sum_{j=0}^{t-1} y_{i-4j} \prod_{k=j+1}^t \overline{y_{i-4k+1}} \overline{y_{i-4k+2}} \overline{y_{i-4k+3}}$
10	$y_i = x_i + \overline{x_{i-1}} \overline{x_{i+1}x_{i+2}} \overline{x_{i+4}}$	$y_{i-4t} + \sum_{j=0}^{t-1} y_{i-4j} \prod_{k=j+1}^t \overline{y_{i-4k-1}} \overline{y_{i-4k+1}} \overline{y_{i-4k+2}} + \prod_{k=1}^t \overline{y_{i-4k-1}} \overline{y_{i-4k+1}} \overline{y_{i-4k+2}}$

Table 2: Details of p_t . The formula of the corresponding inverse can be obtained at $p_{t'}$ where $\lambda \cdot t' = 1 \pmod n$ and the corresponding λ can be referred to Table 1.

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A Appendix

In Table 1 we provide more examples of SI transformations that can be studied with our method. Note that construction of r and choosing (i, j) for any SI functions are not unique, e.g., we can take $r = y_i y_{i-1}$ and $(i, j) = (i, i-1)$ for $y_i = x_i + \overline{x_{i-1}} \overline{x_{i+1}x_{i+2}}$.

Remark 3. Using the parameters given in Table 1, we can easily derive the recursive relation for p_t . By recursively expanding p_t , we can obtain an explicit expression for p_t in terms of y_i 's. The details are given in Table 2.

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